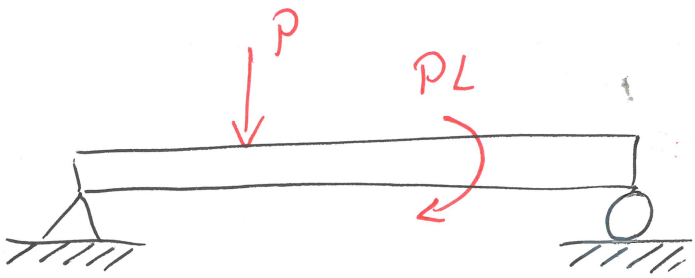
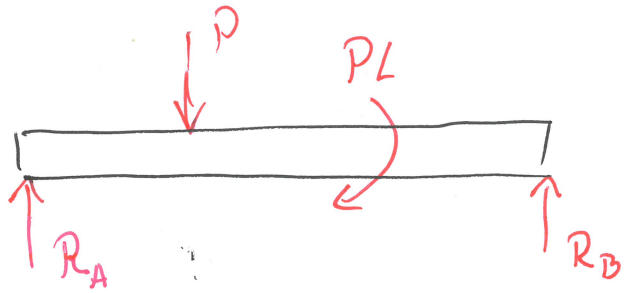
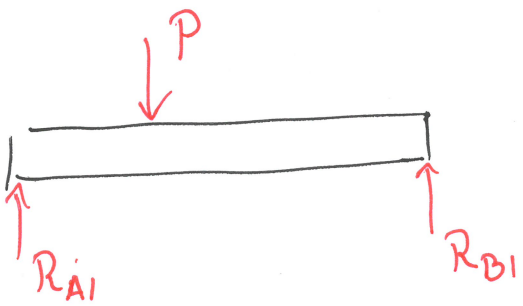
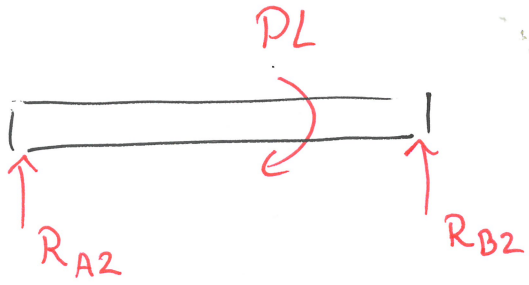
Övning 10 - Elementarfall för böjning

2.4: 102, 103, 120, 122









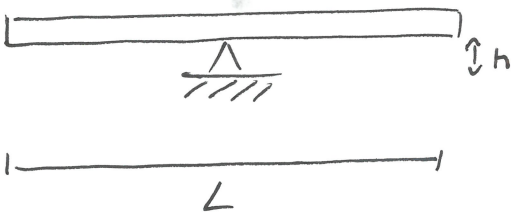
$$R_A = R_{A1} + R_{A2} \qquad R_B = R_{B1} + R_{B2}$$

$$\delta = \delta_1 + \delta_2 \qquad \theta_A = \theta_{A1} + \theta_{A2}$$

$$\theta_B = \theta_{B1} + \theta_{B2}$$

2.4.102

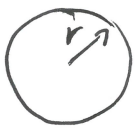
Stång deformeras av egenvikt.



Hur lång får stången vara så ändarna inte släpar i marken?

Sökt: L

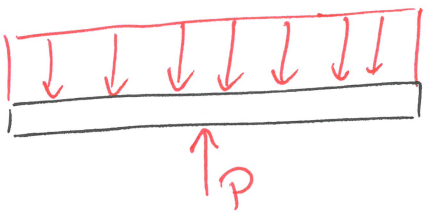
Tvärsnitt



$r = 4\,mm \quad E = 200\,GPa \quad \rho = 7850\,kg/m^3$

$h = 1400\,mm$

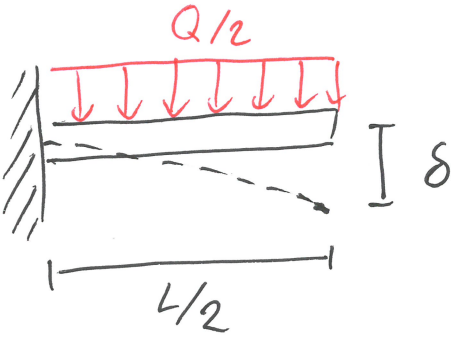
① Frilägg



$$Q = mg = \rho g V = \rho g \pi r^2 L$$

② Skriv som ett elementarfall

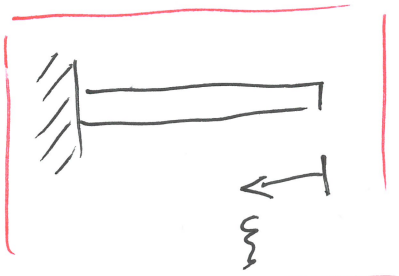
- Problemet helt symmetriskt



[FS. 31.1.3]

$$\delta(\xi) = \frac{\left(\frac{Q}{2}\right)\left(\frac{L}{2}\right)^3}{24EI}\left(\xi^4 - 4\xi + 3\right)$$

ξ böjar i ytterändan!



$$\delta(0) = h = \frac{3\left(\frac{Q}{2}\right)\left(\frac{L}{2}\right)^3}{24EI}$$

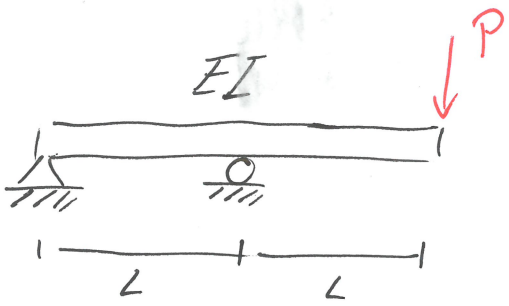
$$h = \frac{3}{24EI}\left(\frac{L}{2}\right)^3\left(\frac{\rho g \pi r^2 L}{2}\right)$$

$$L = \sqrt[4]{\frac{32hEr^2}{\rho g}} \Rightarrow \boxed{L = 6.6\,m}$$

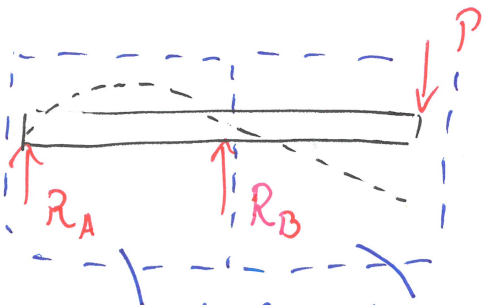
$$I = \frac{\pi r^4}{4} \quad [FS\ 30.1.3]$$

2.4.103

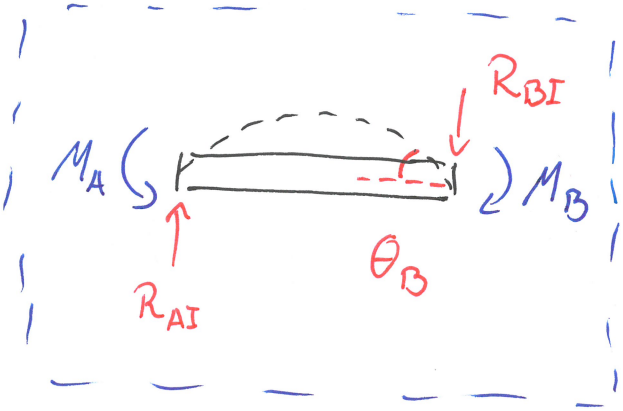
Sökt: Nedböjningen av högra änden δ_C

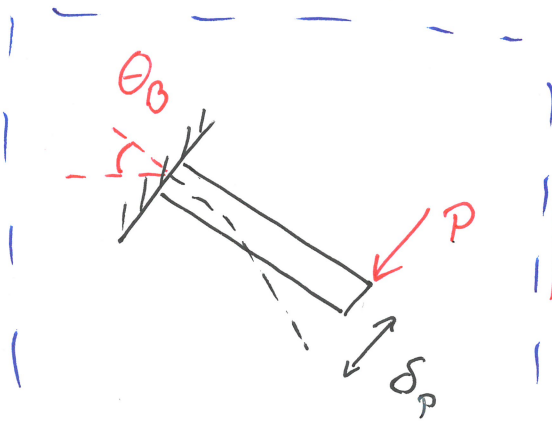


Dela upp problemet + Frilägging



Elementarfall 31.2.5 Elementarfall 31.1.1





$\delta_C = \delta + L \sin\theta_B$

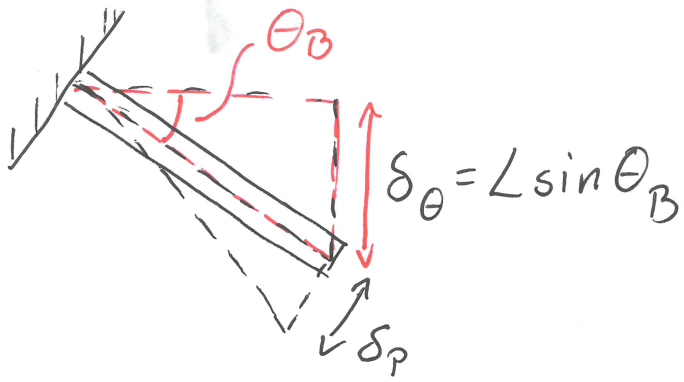
Beräkna θ_B från elementarfall 31.2.5

$$\theta_B = \frac{M_A L}{6EI} + \frac{M_B L}{3EI}$$

($M_A = 0$ $M_B = PL$)
Från frilägging

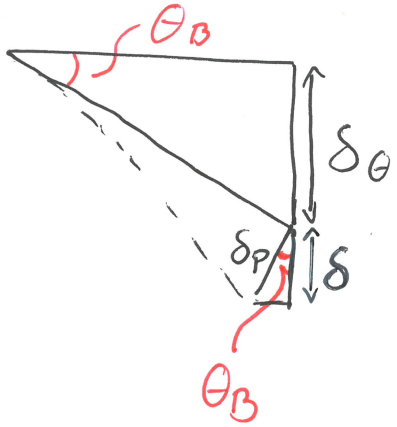
$$\theta_B = \frac{PL^2}{3EI}$$

2.4.103



För små deformationer

$\sin\theta_B \approx \theta_B \qquad \delta_p \approx \delta \quad (\cos\theta_B \approx 1)$



$$\boxed{\delta_c = \delta + \delta_\theta}$$

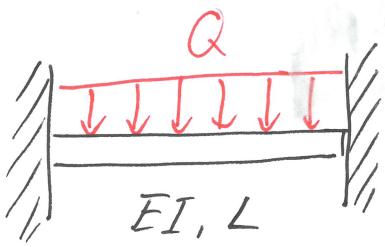
Beräkna δ_p från elementarfall 31.1.1

$$\delta_p = \delta(\alpha = 0, \xi = 0) = \frac{PL^3}{3EI}$$

Beräkna δ_c

$$\delta_c = \frac{PL^3}{3EI} + \left(\frac{PL^2}{3EI}\right)L \Rightarrow \qquad \boxed{\delta_c = \frac{2PL^3}{3EI}}$$

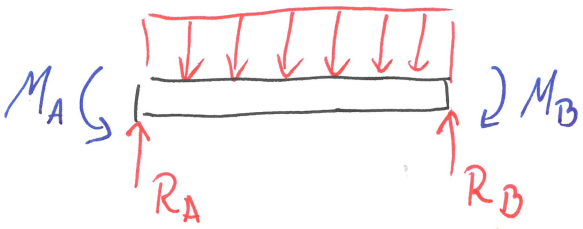
2.4.120



Sökt: Reaktionsmomenten
M_A M_B
Nedböjningen i mitten
$s(\frac{L}{2})$

① Frilägging

symmetriskt

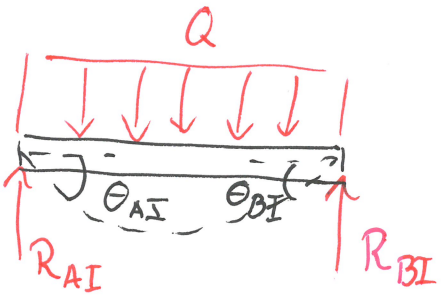


$\uparrow: 2R - Q = 0$

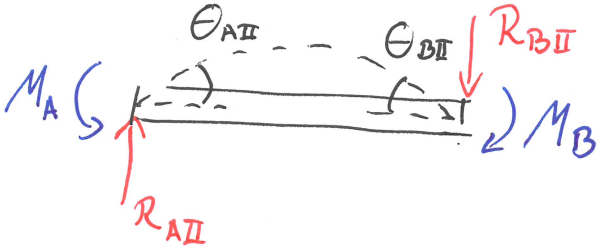
$R = \frac{Q}{2}$

② Kombinera elementarfall

FS 31.2.3



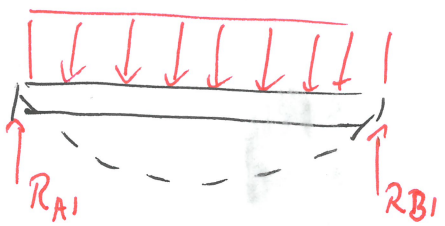
FS 31.2.5



Fast inspänt

$\theta_A = \theta_{AI} - \theta_{AII} = 0$

$\theta_B = \theta_{BI} - \theta_{BII} = 0$

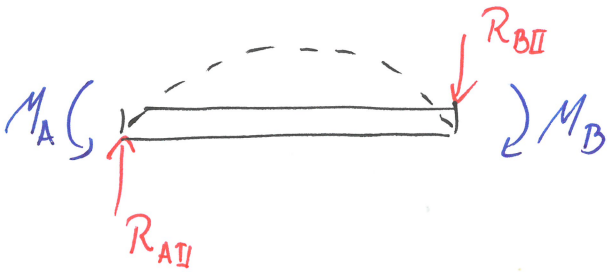


Från FS

$$R_{AI} = R_{BI} = \frac{Q}{2}$$

$$\theta_{AI} = \theta_{BI} = \frac{QL^2}{24EI}$$

$$\delta_I\left(\frac{1}{2}\right) = \frac{5}{384}\frac{QL^3}{EI}$$



Från FS

$$R_{AII} = R_{BII} = \frac{M_A - M_B}{L}$$

$$\theta_{AII} = \frac{M_A L}{3EI} + \frac{M_B L}{6EI}$$

$$\theta_{BII} = \frac{M_A L}{6EI} + \frac{M_B L}{3EI}$$

$$\delta_{II}\left(\frac{1}{2}\right) = \frac{L^2}{6EI}\left(M_A\frac{3}{8} + M_B\frac{3}{8}\right)$$

Fast inspänt

$$\theta_{AI} = \theta_{AII} \Rightarrow \frac{M_A L}{3EI} + \frac{M_B L}{3EI} = \frac{QL^2}{24EI}$$

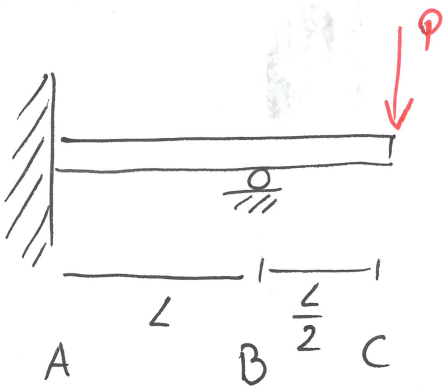
$$\theta_{BI} = \theta_{BII} \Rightarrow \frac{M_A L}{6EI} + \frac{M_B L}{3EI} = \frac{QL^2}{24EI}$$

$$\Rightarrow \boxed{\begin{aligned} M_A &= \frac{QL}{12} \\ M_B &= \frac{QL}{12} \end{aligned}}$$

Totala nedböjning

$$\delta_{tot} = \delta_I\left(\frac{1}{2}\right) - \delta_{II}\left(\frac{1}{2}\right) = \frac{5}{384}\frac{QL^3}{EI} - \frac{L^2}{6EI}\frac{3}{4}\frac{QL}{12} \Rightarrow \boxed{\delta_{tot} = \frac{QL^3}{384EI}}$$

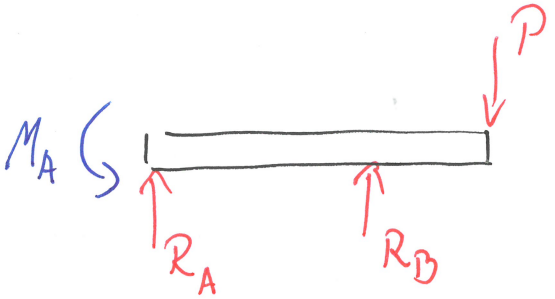
2.4.122



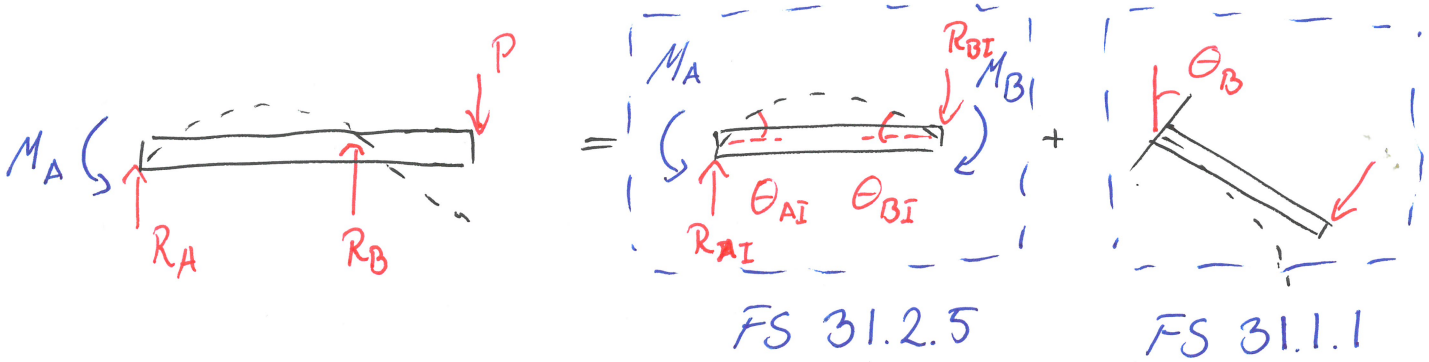
Sökt: Reaktionskraft vid stödet R_B

Utböjningen längst ut till höger δ_C

① Frilägg



② Inget elementarfall – Dela upp



FS 31.2.5 FS 31.1.1

$M_B = PL/2$

FS 31.2.5

Fast inspänt vid A $\theta_{AI} = 0$

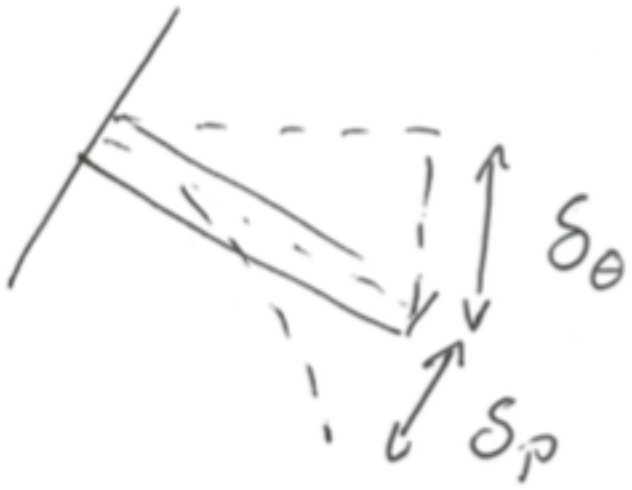
$$\theta_{AI} = \frac{M_A L}{3EI} + \frac{PL^2}{2 \cdot 6EI} = 0 \quad \Rightarrow \quad M_A = -\frac{PL}{4}$$

Eftersom vi har M_A och M_B räkna ut Θ_{BI}

$$\Theta_{BI} = -\frac{PL}{4}\frac{L}{6EI} + \frac{PL}{2}\frac{L}{3EI} = \frac{PL^2}{8EI}$$

FS 31.1.1 $(\beta=1 \quad \alpha=0)$

$$\delta_P = \frac{P(\frac{L}{2})^3}{6EI}(-1+3(1-0)) = \frac{PL^3}{24EI}$$

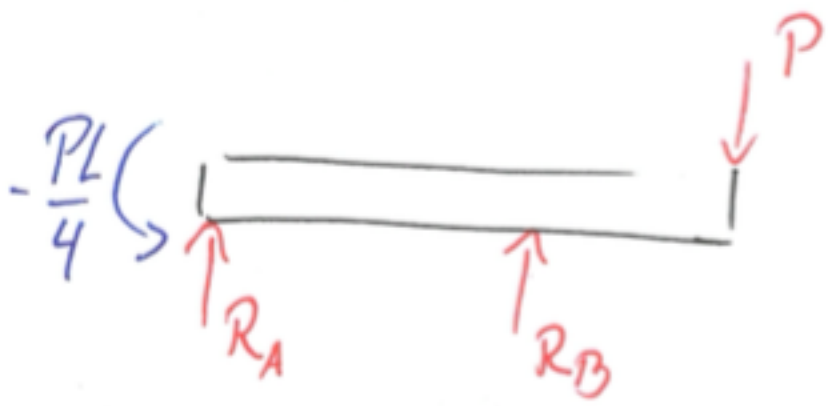


$$\delta_{tot} = \delta_\theta + \delta_P$$

$$\delta_\theta = \sin\theta_B \cdot (\tfrac{L}{2}) \approx \frac{L\theta_B}{2} = \frac{PL^3}{16EI}$$

$$\boxed{\delta_{tot} = \frac{PL^3}{16EI} + \frac{PL^3}{24EI} = \frac{5PL^3}{48EI}}$$

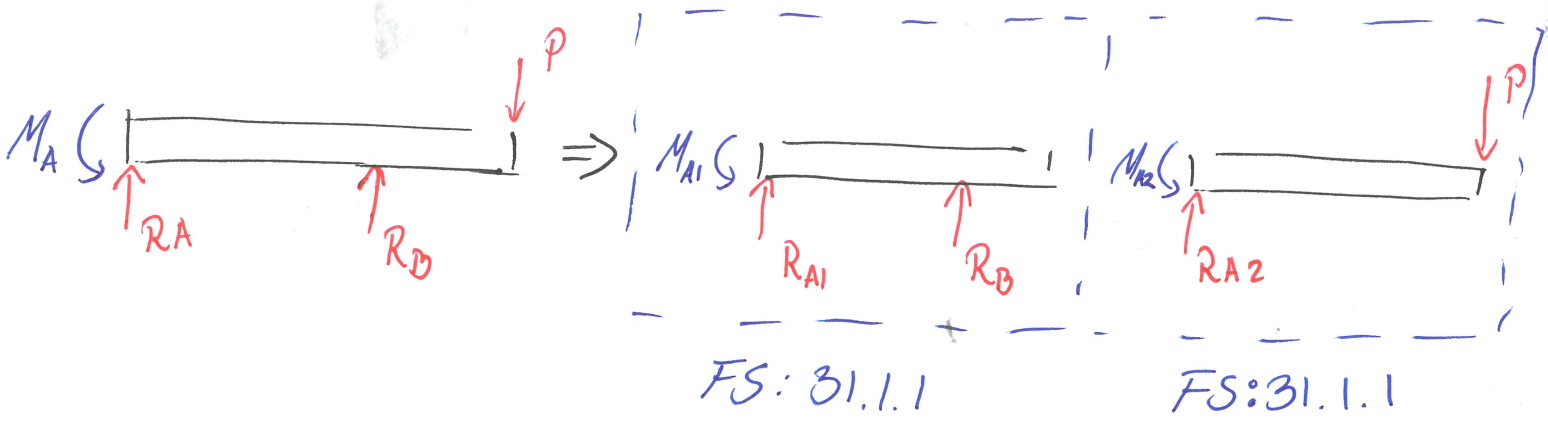
Reaktionskraften R_B kan nu bestämmas



$$\circlearrowleft A \quad -\left(-\frac{PL}{4}\right) - R_B L + P\frac{3}{2}L = 0$$

$$\boxed{R_B = \frac{7}{4}P}$$

2.4.122 Alternativ 2



Krav: $\delta_B = \delta_{B1} + \delta_{B2} = 0$ Sökt: $\delta_C = \delta_{C1} + \delta_{C2}$

① $$\delta_{B1} = \delta\left(\alpha = \frac{1}{3}, \xi = \frac{1}{3}\right) = \frac{-R_B\left(\frac{3L}{2}\right)^3}{3EI}\left(\frac{2}{3}\right)^3 = -\frac{R_B L^3}{3EI}$$

$$\delta_{C1} = \delta\left(\alpha = \frac{1}{3}, \xi = 0\right) = -\frac{R_B\left(\frac{3L}{2}\right)^3}{6EI}\left(-\left(\frac{2}{3}\right)^3 + 3\left(\frac{2}{3}\right)^2\right)$$

$$\delta_{C1} = -\frac{7}{12}\frac{R_B L^3}{EI}$$

② $$\delta_{B2} = \delta\left(\alpha = 0, \xi = \frac{1}{3}\right) = \frac{P\left(\frac{3L}{2}\right)^3}{6EI}\left(\left(\frac{1}{3}\right)^3 - 3\left(\frac{1}{3}\right) + 2\right)$$

$\xi > \alpha$

$$\delta_{B2} = \frac{7PL^3}{12EI}$$

$$\delta_{C2} = \delta(\alpha = 0, \xi = 0) = \frac{P\left(\frac{3L}{2}\right)^3}{3EI} = \frac{9PL^3}{8EI}$$

$$\delta_{B1} + \delta_{B2} = -\frac{R_B L^3}{3EI} + \frac{7PL^3}{12EI} = 0 \Rightarrow \boxed{R_B = \frac{7}{4}P}$$

$$\delta_C = \delta_{C1} + \delta_{C2} = -\frac{7}{12}\frac{7}{4}\frac{PL^3}{EI} + \frac{9}{8}\frac{PL^3}{EI} = \boxed{\frac{5PL^3}{48EI}}$$